Bayesian Network

A Bayesian network falls under the category of probabilistic graphical modelling (PGM) technique that is used to compute uncertainties by using the concept of probability.

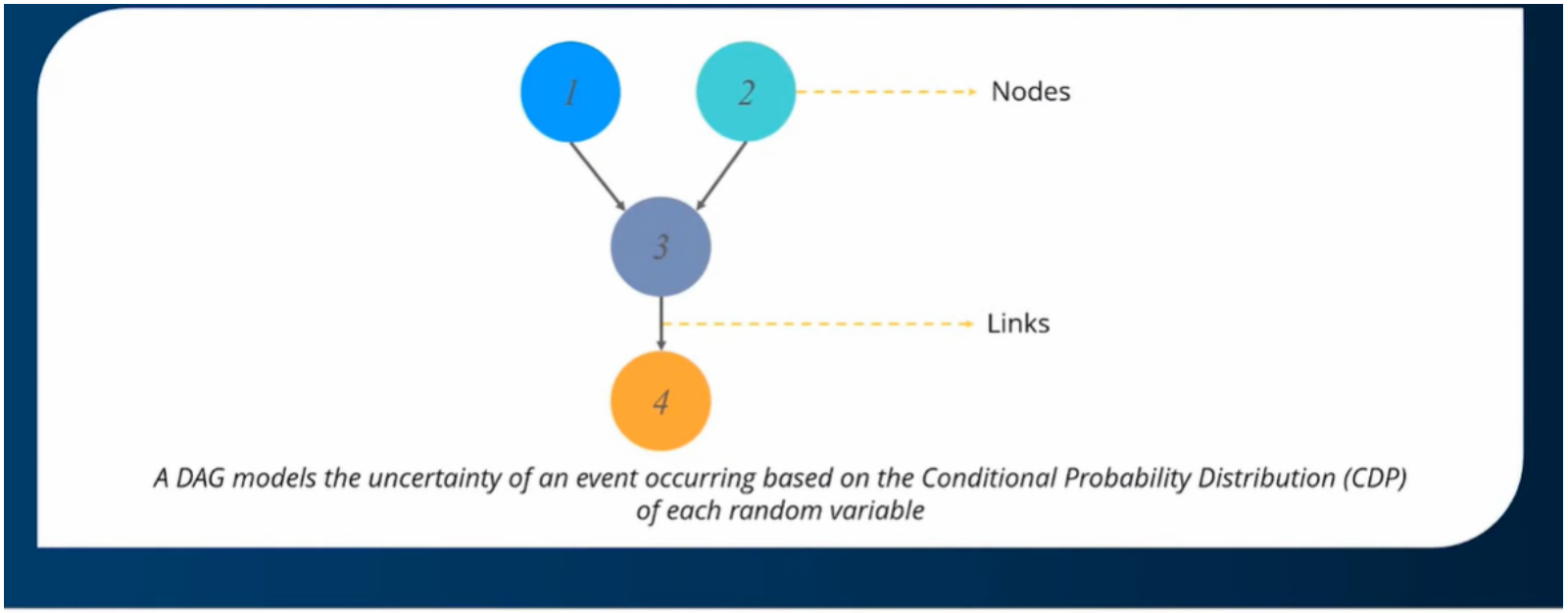
It is also called a **Bayes network, belief network, decision network**, or **Bayesian model**.

Bayesian network is key computer technology for dealing with probabilistic events and to solve a problem which has uncertainty. We can define a Bayesian network as:

"A Bayesian network is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph."

Bayesian network is used to model the uncertainties (unknown values) by using directed acyclic graph.

In a nutshell we can say Bayesian network is based on the concept of probability and it can be represented by using a directed acyclic graph.



What Is A Directed Acyclic Graph?

A Directed Acyclic Graph is used to represent a Bayesian Network and like any other statistical graph, a DAG contains a set of nodes and links, where the links denote the relationship between the nodes.

In computing terms, node represents the random variables and the edges defines the relationship between these variables.

But the question arise that what output you can get from the directed acyclic graph?

It basically tells you the uncertainty of an event occurring based on the conditional probability distribution of each random variable.

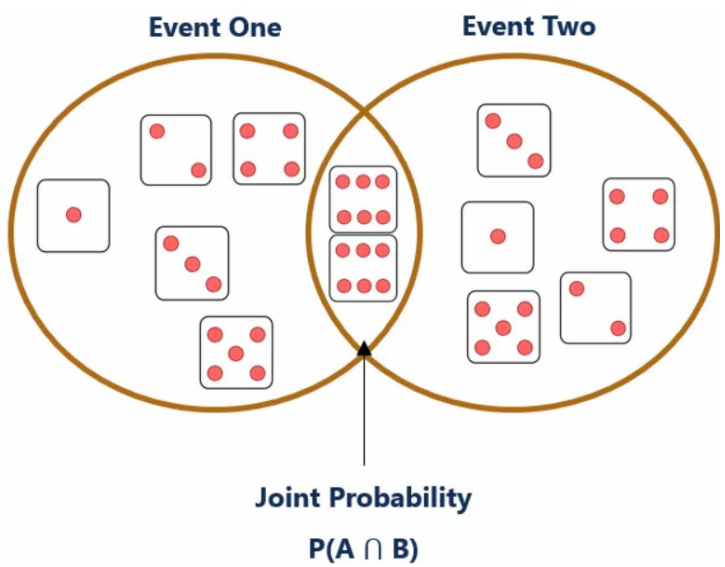
Math Behind Bayesian Networks

As mentioned earlier, Bayesian models are based on the simple concept of probability. So let's understand what conditional probability and Joint probability distribution mean.

Joint Probability: A joint probability, in probability theory, refers to the probability that two events will both occur. In other words, joint probability is the likelihood of two events occurring together.

…….it is a measure of two events happening at the same time, i.e., p (A and B).

Formula: $P(A \cap B) = P(A) \times P(B)$



A joint probability can be visually represented through a Venn diagram. Consider the joint probability of rolling two 6's in a fair six-sided dice

Shown on the Venn diagram above, the joint probability is where both circles overlap each other. It is called the "intersection of two events."

For example:

Ques 1: What is the probability of rolling the number five twice in a fair six-sided dice?

Ques 2: What is the probability of getting a head followed by a tail in a coin?

Conditional Probability: of an event B is the probability that the event will occur given that an event A has already occurred.

P(B|A) = probability of event B occurring, given that event A occurs.

If A and B are dependent events: $P(B|A) = P(A \text{ and } B)/P(A)$

If A and B are independent events: $P(B|A) = P(B)$

Example: Given that you drew a red card, what's the probability that it's a four?

Solution:

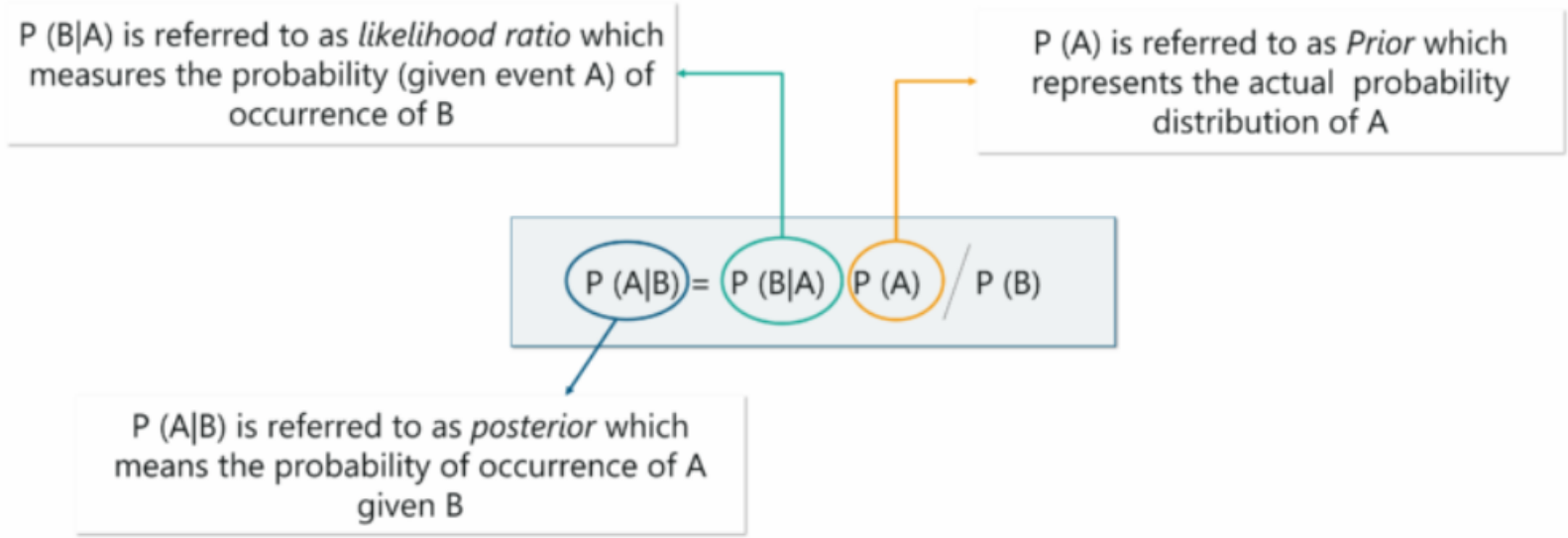
$(p(\text{four}|\text{red}))=2/26=1/13$.

So out of the 26 red cards (given a red card), there are two fours so $2/26=1/13$.

Bayes' Theorem

The Bayes theorem is used to calculate the conditional probability, which is nothing but the probability of an event occurring based on prior knowledge of conditions that might be related to the event.

Mathematically, the Bayes theorem is represented as:



In the above equation:

P(A|B): Conditional probability of event A occurring, given the event B

P(A): Probability of event A occurring

P(B): Probability of event B occurring

P(B|A): Conditional probability of event B occurring, given the event A

Formally, the terminologies of the Bayesian Theorem are as follows:

A is known as the proposition and B is the evidence

P(A) represents the prior probability of the proposition

P(B) represents the prior probability of evidence

P(A|B) is called the posterior

P(B|A) is the likelihood

Therefore, the Bayes theorem can be summed up as:

Posterior=(Likelihood) (Proposition prior probability)/Evidence prior probability

Bayesian Networks Example

Let's assume we are creating a Bayesian network that will model the marks (m) of a student on his examination. So obviously marks depend on couple of factors.

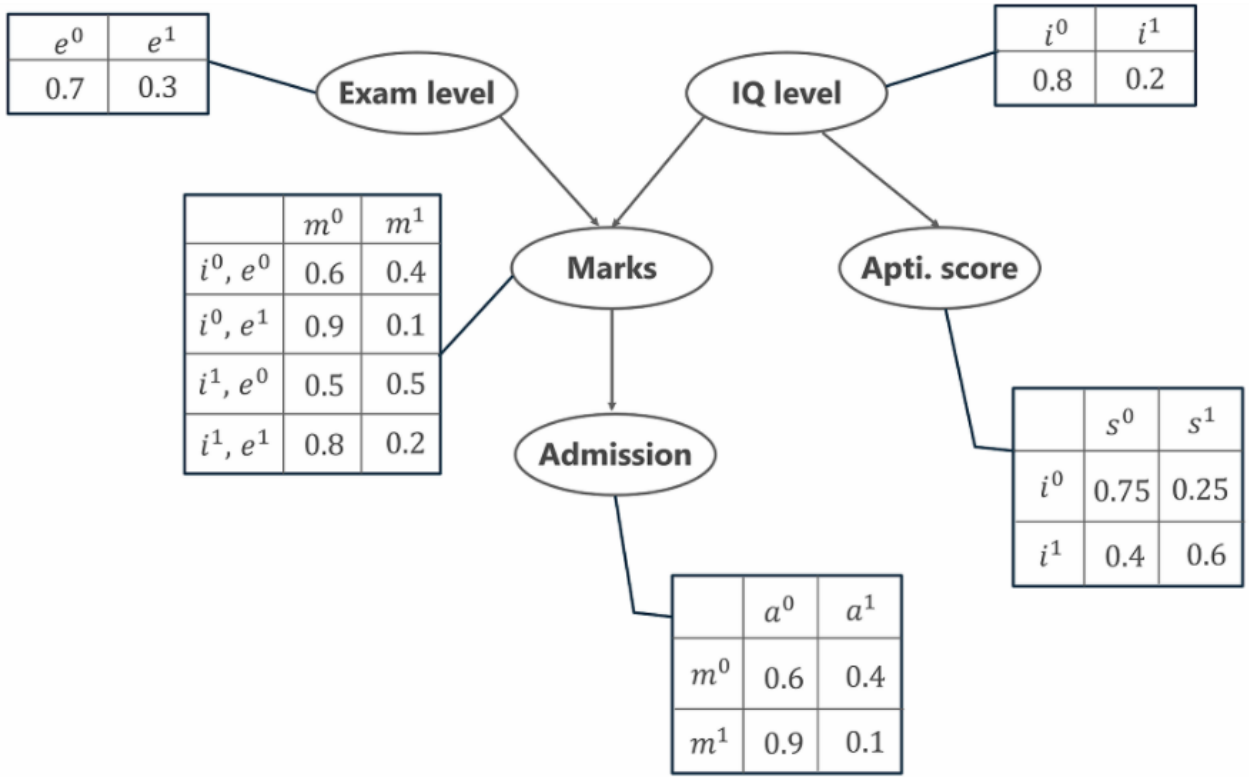
- Exam level (e): This is a discrete variable that can take two values, (difficult, easy)
- IQ of the student (i): A discrete variable that can take two values (high, low)

- The marks will intern predict whether or not he/she will get admitted (a) to a university.
- The IQ will also predict the aptitude score (s) of the student.

With this information, we can build a Bayesian Network that will model the performance of a student on an exam. The Bayesian Network can be represented as a DAG where each node denotes a variable that predicts the performance of the student.

Create a Bayesian Network that will model the marks (m) of a student on his examination

Conditional Probability Table



The marks will depend on:

- Exam level (e): (difficult, easy)
- IQ of the student (i): (high, low)
- Marks → admitted (a) to a university
- The IQ → aptitude score (s) of the student

Steps:

1. Marks will depend exam level which is denoted by e. Exam level is basically a discrete variable that can take two values. So, the exam can either be difficult or it can be easy.
2. If you look at the figure, you can see that the exam level is directly connected to the marks. Meaning that your marks are dependent on your exam level.
3. Next we have IQ of the student. Now IQ of student is again a discrete variable that can take two values. It has value high and low. So your marks are also directly dependent on your IQ level.
4. Now the marks will in turn predict whether or not a student will get admitted into a University.
5. Now this admission into a university is represented by a variable (a).

So the probability of a person getting admitted into a university will depend on marks. That is why there is direct link of marks and admission.

IQ level will also produce the aptitude score of a student. Meaning that your aptitude score will directly dependent on your IQ level.

So represented this distribution through a directed acyclic graph and conditional probability table. Right you can see the table that is connected to each of these nodes these are basically the conditional probability of each of these nodes. For exam level and IQ level since they are root nodes or since they have no any parent node they have just the basic probability.

But, when it comes to marks, you can see that conditional probability table has separate column for IQ level and for exam level. Right, because the marks is directly dependent on exam level and IQ level and table basically represent the conditional probability.

Similarly, for aptitude score, you can see that the conditional probability table has a separate column for IQ and for admission there is a separate column for marks. This is because there are links between these nodes.

Right I hope all of you understood that how **can you represent a problem statement through a directed acyclic graph. This is the first step of building the Bayesian Network.**

We have already discussed that this graph is clearly show that how each variable or each node is directly dependent on its parent node.

Since we are creating the Bayesian Network that will model the marks (m) of a student on his examination based on his performance.

So far that we are going to factorizing the joint probability distribution. As we have discussed that joint probability distribution deals with the probability of events occurring at the same time.

The formula of joint probability distribution

$$P(a, m, i, e, s) = P(a \mid m)\, P(m \mid i, e)\, p(i)\, P(e)\, P(s \mid i)$$

Here,

P (a | m) represents the conditional probability of a student getting an admission based on his marks: Conditional Probability (CP) of student admit -> marks.

P (m | i, e) represents the conditional probability of the student's marks, given his IQ level and exam level: CP of the student's marks -> (IQ and exam level)

P (i) denotes the probability of his IQ level (high or low)

P (e) denotes the probability of the exam level (difficult or easy)

P (s | i) denotes the conditional probability of his aptitude scores, given his IQ level: aptitude scores - > IQ level

We have studied the relationship between the parent and the child node and represented with directed acyclic graph and be then measured the joint probability distribution.

Now if you have notice already that is a pattern here that the probability of a random variable is always depends on its parent nodes. Therefore, we formulate the Bayesian Network as:

$$P(X_1, \dots, X_n) = \Pi_{i=1}^{n} p(X_i \mid Parents(X_i))$$

Here Xi basically represents the random variable whose probability depends on the probability of his parent nodes.

This is the thing about the Bayesian network. This is one of the simplest and yet the most effective techniques that are applied in predictive modeling, descriptive analysis and so on.

Bayesian Networks are used in such cases that involve predicting uncertain tasks and outcomes. In the below section you'll understand how Bayesian Networks can be used to solve more such problems.

Bayesian Networks Application

Bayesian Networks have innumerable applications in a varied range of fields including healthcare, medicine, bioinformatics, information retrieval and so on. Here's a list of real-world applications of the Bayesian Network:

1. **Disease Diagnosis:** Bayesian Networks are commonly used in the field of medicine for the detection and prevention of diseases. They can be used to model the possible symptoms and predict whether or not a person is diseased.

2. **Optimized Web Search:** Bayesian Networks are used to improve search accuracy by understanding the intent of a search and providing the most relevant search results. They can effectively map users' intent to the relevant content and deliver the search results.

3. **Spam Filtering:** Bayesian models have been used in the Gmail spam filtering algorithm for years now. They can effectively classify documents by understanding the

contextual meaning of a mail. They are also used in other document classification applications.

4. **Gene Regulatory Networks:** GRNs are a network of genes that are comprised of many DNA segments. They are effectively used to communicate with other segments of a cell either directly or indirectly. Mathematical models such as Bayesian Networks are used to model such cell behavior in order to form predictions.

5. **Biomonitoring:** Bayesian Networks play an important role in monitoring the quantity of chemical dozes used in pharmaceutical drugs.